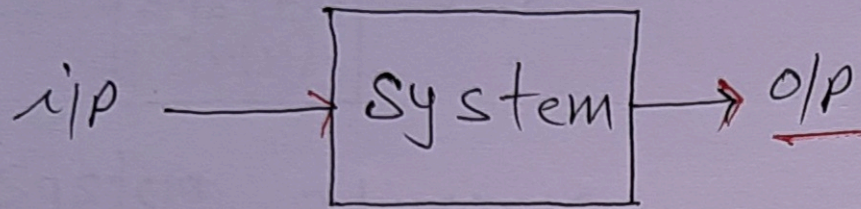


# Response of System



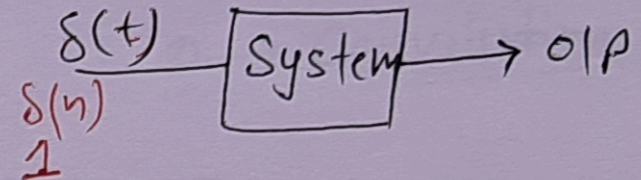
Methods to find Response

- ① Convolution Method (Graphical Method)
- ② Fourier Transform Method
- ③ Laplace Transform Method
- ④ Z-Transform Method

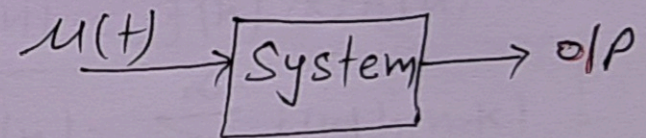
$$x(t) \xrightarrow{\text{Sys}} y(t)$$

$$\frac{Y(s)}{X(s)} = \text{System T/f function}$$

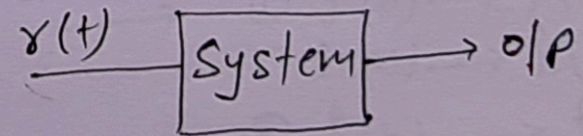
i) Unit impulse Response



ii) Unit step Response



iii) Ramp Response

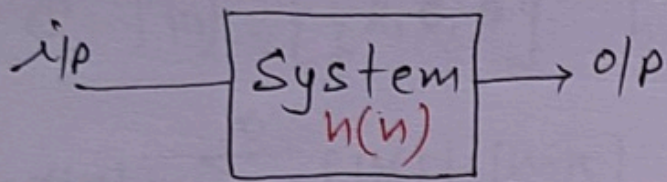


Note:- Unit impulse Response is known as System Response

$$\text{o/p} = \text{System Response} \times \text{input}$$



# O/p Response of Discrete Time Signal



System  $\rightarrow$  i) linear  $\checkmark$   
LTI, ii) Time invariant  $\checkmark$

It means  $\rightarrow$

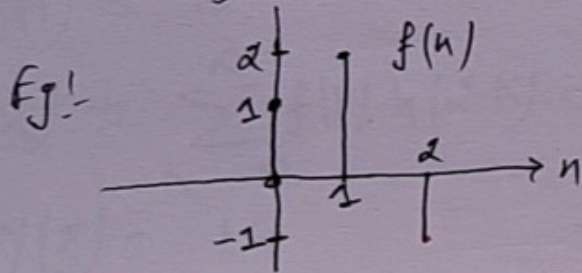
$$\delta(n) \longrightarrow h(n)$$

$$a\delta(n) \longrightarrow ah(n)$$

$$\delta(n-n_1) \longrightarrow h(n-n_1)$$

$$+ \left[ a\delta(n-n_1) \longrightarrow ah(n-n_1) \right]$$

$$b\delta(n-n_2) \longrightarrow bh(n-n_2)$$



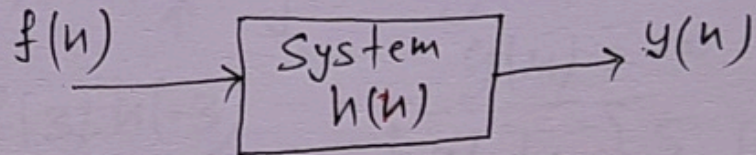
$$f(n) = \delta(n) + 2\delta(n-1) - \delta(n-2)$$

input signal = must be  $\sum$  impulse signals.

for any input  $f(n)$

$$\text{Response } y(n) = f(n) \otimes h(n)$$

$\otimes \rightarrow$  Convolution



$$y(n) = f(n) \otimes h(n)$$

$$y(n) = \sum_{k=-\infty}^{\infty} f(k)h(n-k)$$

- operation  $\rightarrow$
- ①  $f(n) \rightarrow f(k)$   
 $h(n) \rightarrow h(k)$
  - ②  $h(k) \rightarrow h(-k)$
  - ③  $h(-k) \rightarrow h(-k+n)$

Convolution Method  $\rightarrow$  Graphical Procedure



# Problem of Convolution Method

①  $f(n) = \{ \underset{\uparrow}{1}, 2, 0, 0 \}$ ,  $h(n) = \{ \underset{\uparrow}{5}, 6, 7, 0 \}$  find the Response

$$y(n) = \sum_{k=-\infty}^{\infty} f(k) h(n-k)$$

$$f(k) = \{ \underset{\uparrow}{1}, 2, 0, 0 \}$$

$$h(k) = \{ \underset{\uparrow}{5}, 6, 7, 0 \}$$

$$h(-k) = \{ 0, 7, 6, \underset{\uparrow}{5} \}$$

$$y(0) = 5$$

$$y(1) = 16$$

$$y(2) = 19$$

$$y(3) = 14$$

$$y(4) = 0$$

$$y(0) = \sum_{k=-\infty}^{\infty} f(k) h(-k)$$

$$= f(0)h(0) + f(1)h(-1) + f(2)h(-2) + f(3)h(-3)$$

$$= 1 \times 5 + 2 \times 0 + 0 \times 0 + 0 \times 0$$

$$y(0) = 5$$

$$y(1) = \sum_{k=-\infty}^{\infty} f(k) h(1-k) = f(0)h(1) + f(1)h(0) + f(2)h(-1) + f(3)h(-2)$$

$$= 1 \times 6 + 2 \times 5 + 0 \times 0 + 0 \times 0$$

$$y(1) = 16$$

$$= 6 + 10 = 16$$

$$y(2) = \sum_{k=-\infty}^{\infty} f(k) h(2-k) = f(0)h(2) + f(1)h(1) + f(2)h(0) + f(3)h(-1)$$

$$= 1 \times 7 + 2 \times 6 + 0 \times 5 + 0 \times 0$$

$$y(2) = 19$$

$$= 7 + 12 + 0 + 0$$

$$= 19$$

$$y(3) = \sum_{k=-\infty}^{\infty} f(k) h(3-k) = f(0)h(3) + f(1)h(2) + f(2)h(1) + f(3)h(0)$$

$$= 1 \times 0 + 2 \times 7 + 0 \times 6 + 0 \times 5$$

$$= 0 + 14 + 0 + 0 = 14$$

$$y(n) = \{ \underset{\uparrow}{5}, 16, 19, 14, 0 \}$$



# Tabular Method of finding Convolution

①  $f(n) = \{1, 2, 3, 4\}$ ,  $h(n) = \{5, 6, 7, 8\}$  find Response

|        | 5  | 6  | 7  | 8  |
|--------|----|----|----|----|
| $f(n)$ |    |    |    |    |
| 1      | 5  | 6  | 7  | 8  |
| 2      | 10 | 12 | 14 | 16 |
| 3      | 15 | 18 | 21 | 24 |
| 4      | 20 | 24 | 28 | 32 |

$$y(n) = [5, 16, 34, 60, 61, 52, 32]$$

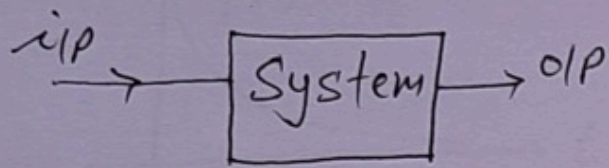
$$f(n) \rightarrow N = 4$$

$$h(n) \rightarrow M = 4$$

$$y(n) \rightarrow \frac{N+M-1}{4+4-1} = \underline{\underline{7}}$$



# Output Response of Continuous time signal using Convolution Method.

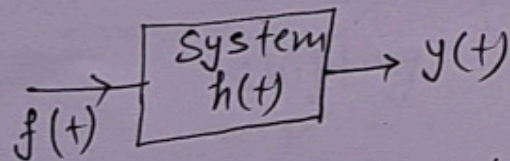


if input  $s(t)$ , o/p  $h(t)$

$h(t)$  impulse Response of system

let  $f(t) \rightarrow$  input signal  
 $h(t) \rightarrow$  impulse Response of system

$y(t) \rightarrow$  o/p Response



$$y(t) = f(t) \otimes h(t)$$

$$y(t) = \int_{-\infty}^{\infty} f(\tau) \cdot h(t-\tau) \cdot d\tau$$

\* System is linear, Time invariant  
"LTI"

## Step to find Convolution

1)  $f(t) \rightarrow f(\tau)$

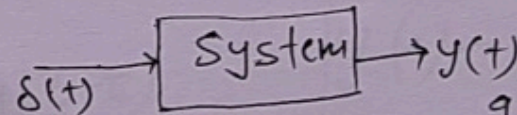
$h(t) \rightarrow h(\tau)$

2)  $h(\tau) \rightarrow h(-\tau)$

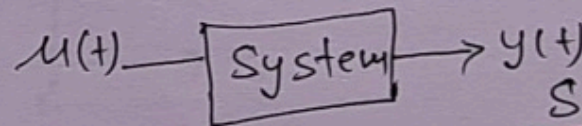
$h(-\tau) \xrightarrow{\text{shift}} h(t-\tau)$  or  $h(-\tau+t)$

3)  $f(\tau) \cdot h(t-\tau) = y(t)$

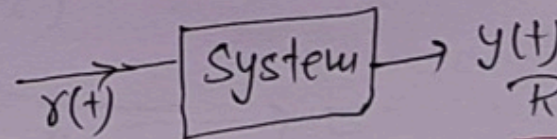
## Graphical Procedure



Impulse Response



Step Response

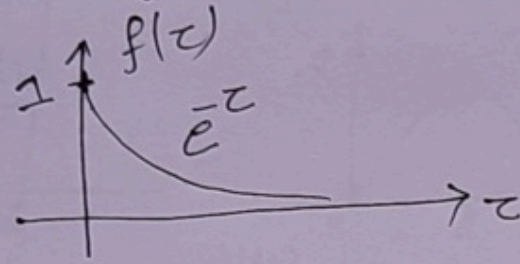
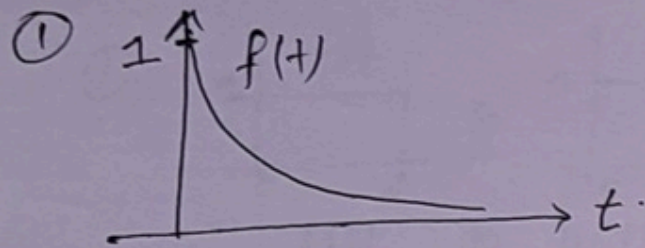


Ramp Response

Note:-  $\frac{d}{dt}(\text{step Res}) = \text{Impulse Response}$   
 $\frac{d}{dt}(\text{ramp Res}) = \text{Step Response}$



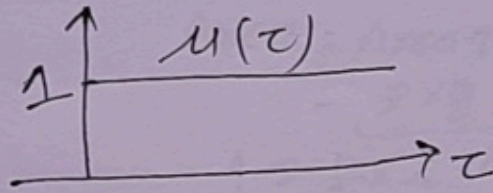
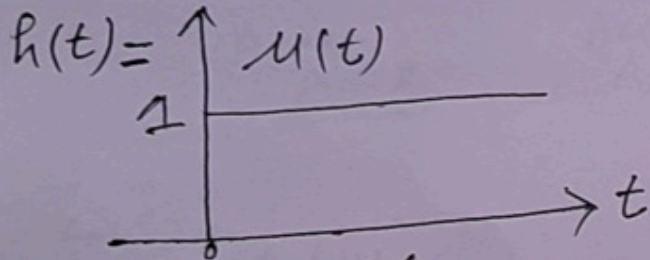
# Problem of convolution method



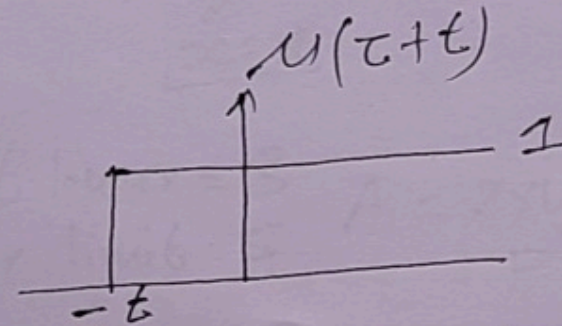
$$y(t) = \int_0^t e^{-\tau} \cdot 1 \cdot d\tau$$

$$y(t) = \int_0^t e^{-\tau} d\tau$$

$$\left. \frac{e^{-\tau}}{-1} \right|_0^t$$

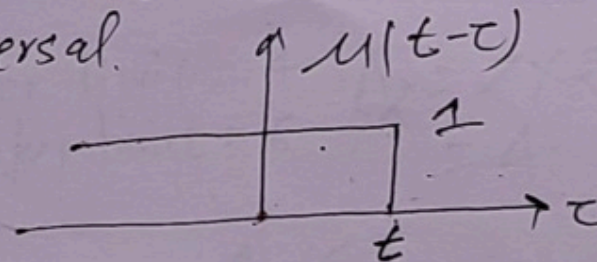


find  $y(t) = \int_{-\infty}^{\infty} f(\tau) \cdot \underline{h(t-\tau)} \cdot d\tau$



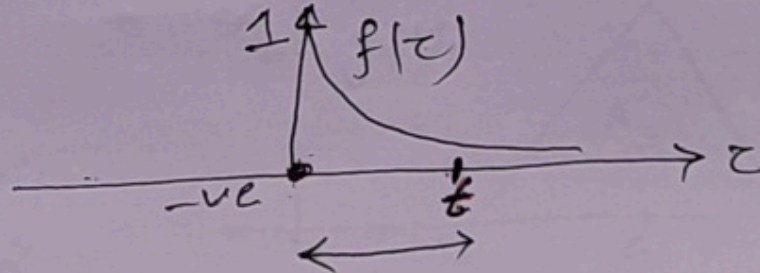
$$y(t) = -[e^{-t} - 1]$$

$u(\tau) \rightarrow u(t-\tau)$   
 shifting  $\checkmark$  Time Reversal  $\checkmark$



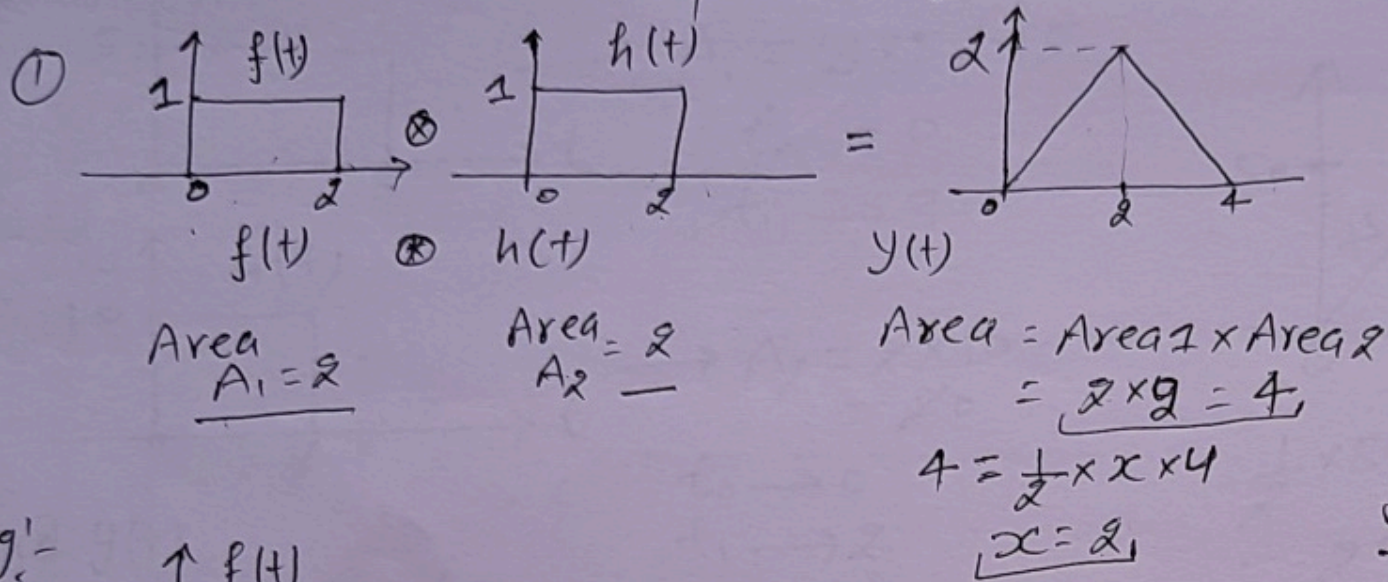
$y(t) = 1 - e^{-t}$

$f(t) \rightarrow \boxed{h(t)} \rightarrow y(t)$   
 $y(t) = f(t) \otimes h(t)$





# Properties of Convolution

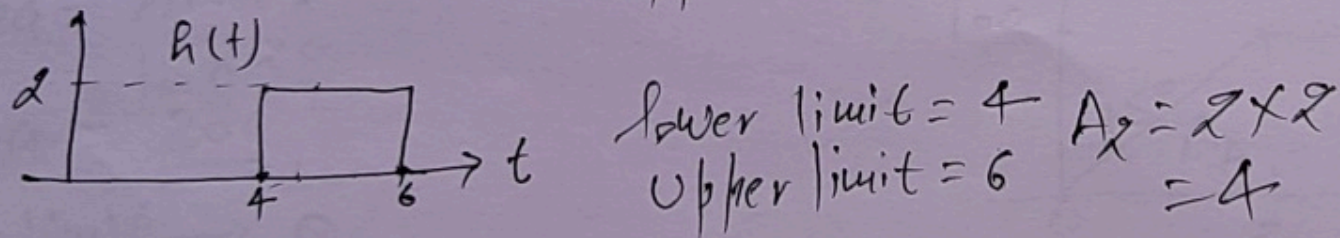
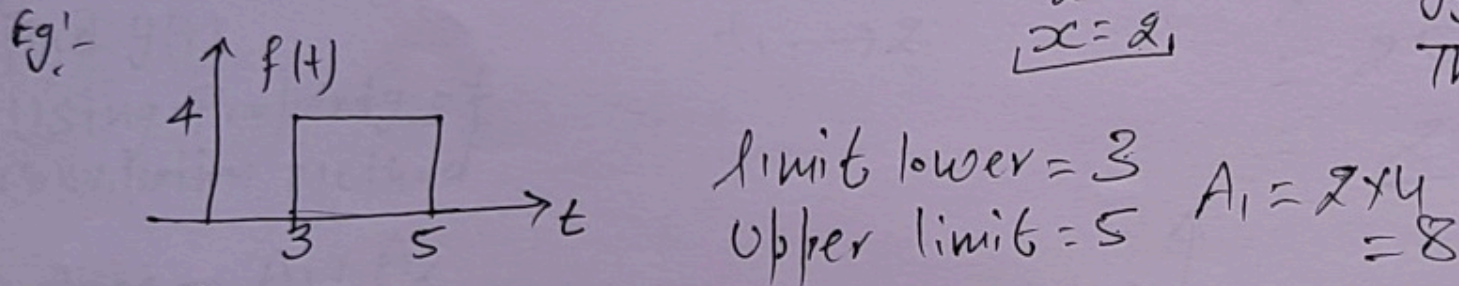


lower limit  $\rightarrow$  lower limit of 1 + lower limit of 2

lower limit  $\rightarrow 0 + 0 = 0$

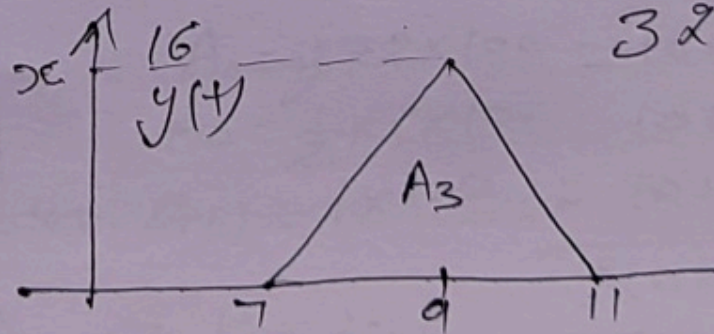
upper limit  $\rightarrow 2 + 2 = 4$

if same width of  $f(t)$ ,  $h(t)$ .  
Then Response Triangular Wave.



find  $y(t)$

lower limit = 7  
 upper limit = 11  
 Total Area =  $8 \times 4$   
 $= 32$

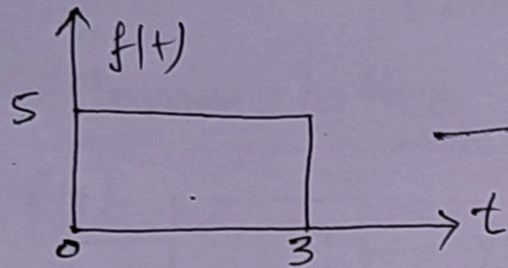


$$32 = \frac{1}{2} \times x \times 4$$

$$\underline{\underline{x = 16}}$$



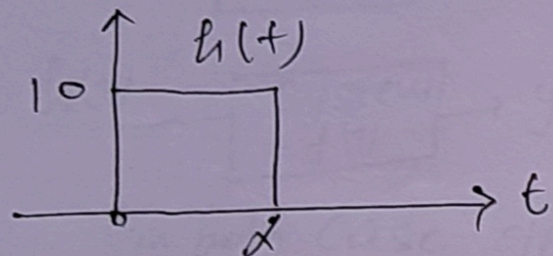
②



$$\rightarrow A_1 = 3 \times 5 = 15$$

$$t_0 \rightarrow 0$$

$$t_1 \rightarrow 3$$



$$\rightarrow A_2 = 2 \times 10 = 20$$

$$t_0 \rightarrow 0$$

$$t_1 \rightarrow 2$$

find  $y(t)$

Using Property of Convolution method.

$$\text{Area} = A_1 \times A_2$$

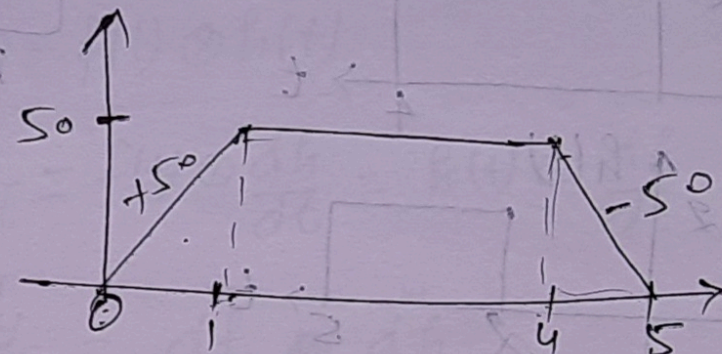
$$= 15 \times 20$$

$$\text{Area} = \underline{\underline{300}}$$

lower limit  $\rightarrow 0$

upper limit  $\rightarrow 5$

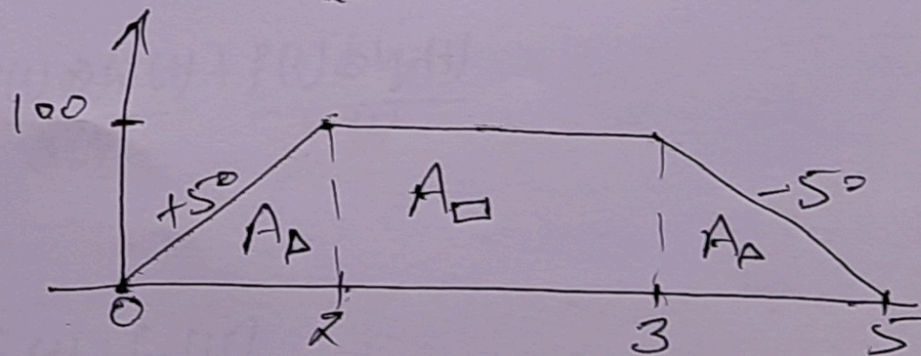
$$\text{slope} = 5 \times 10 = 50$$



$$\frac{1}{2} \times 50 \times 1 + 3 \times 50 + \frac{1}{2} \times 1 \times 50$$

$$= 25 + 150 + 25$$

$$= \underline{\underline{200}}$$



$$A_D = \frac{1}{2} \times 1 \times 100 = 50$$

$$A_Q = 2 \times 100 = 200$$

$$A_D = \frac{1}{2} \times 1 \times 100 = 50$$

$$\underline{\underline{300}}$$

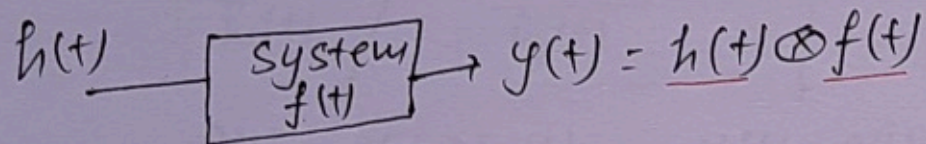
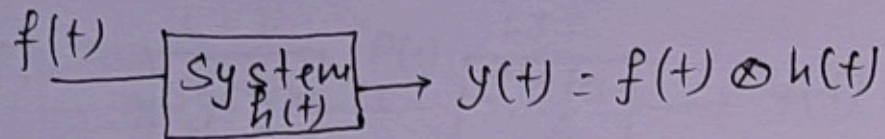
Width same  $\rightarrow$  Traingular  $\triangle$

Width same  $\rightarrow$  Trapezoidal  $\nabla$



# Properties of Convolution Method

② Commutative  $a \times b = b \times a$



in both case o/p remain same.

$$y(t) = f(t) \otimes h(t)$$

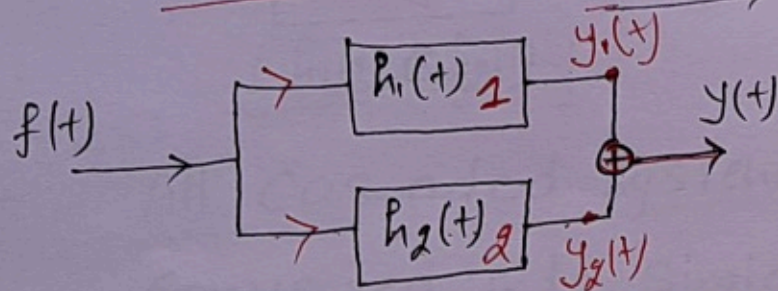
$$\frac{dy}{dt} = f(t) \otimes \frac{dh}{dt} = h(t) \otimes \frac{df}{dt}$$

$$\frac{d^h y}{dt^h} = \frac{d^x f}{dt^x} \otimes \frac{d^m h}{dt^m}$$

where  $\boxed{h = x + m}$

③ Distributive Property

$$a \times (b+c) = a \times b + a \times c$$



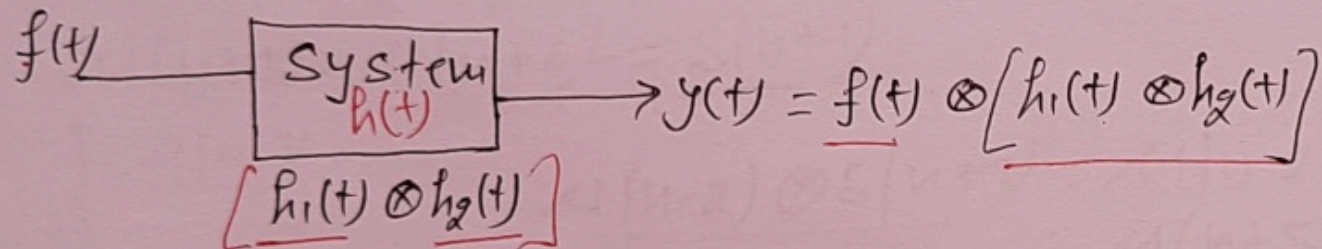
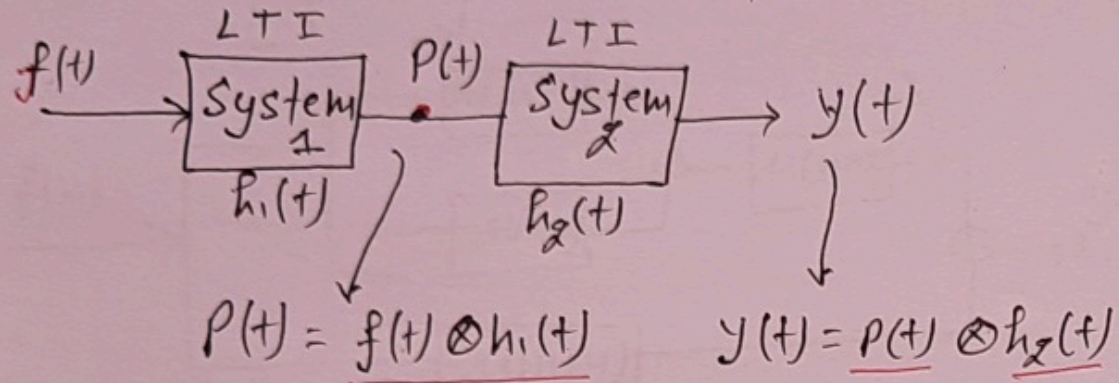
$$y(t) = \frac{f(t) \otimes h_1(t)}{y_1(t)} + \frac{f(t) \otimes h_2(t)}{y_2(t)}$$

$$f(t) \rightarrow \boxed{h_1(t) + h_2(t)}_{h(t)} \rightarrow y(t) = f(t) \otimes [h_1(t) + h_2(t)]$$

\* If multiple LTI System Connected in Parallel  
Then Convert into single LTI System  
whose I.R = sum of all individual I.R.



④ Associative Property  $a \times (b \times c) = (a \times b) \times c$



\* All cascaded system (LTI) can be converted into single system (LTI) whose IR is convolution of all individual impulse response.



# Problem on Property of Convolution method.

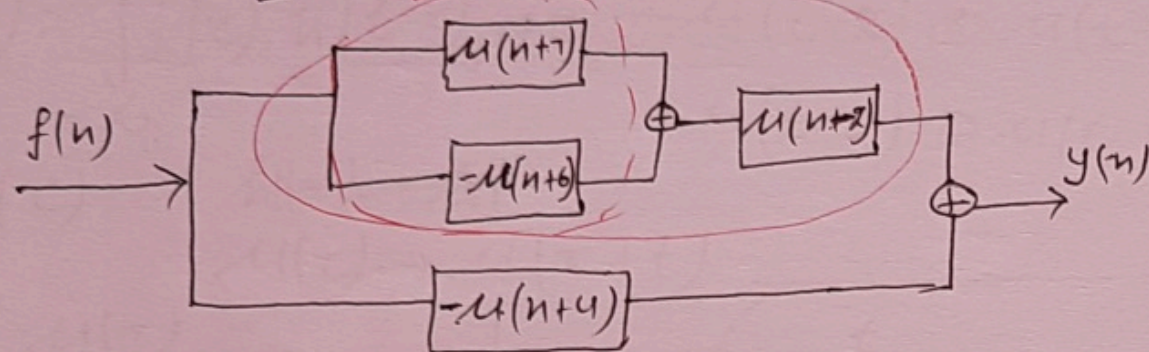
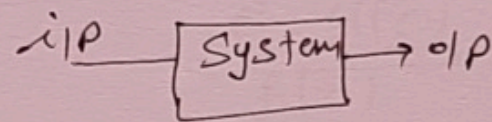
①  $f(n) = \{ \uparrow 2, 3 \}$

Basic Property of  $\delta(n)$

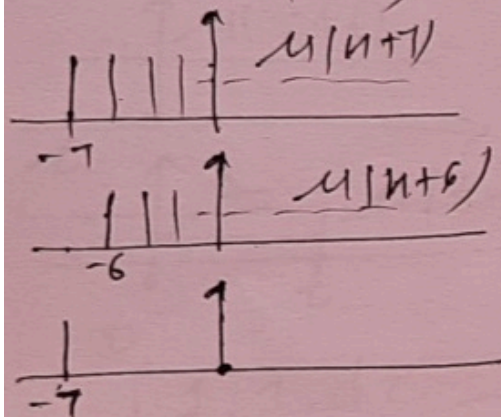
$$f(n) \otimes \delta(n) = f(n)$$

$$f(n) \otimes \delta(n-n_0) = f(n-n_0)$$

$$f(n-n_1) \otimes \delta(n-n_2) = \underline{f(n-n_1-n_2)}$$

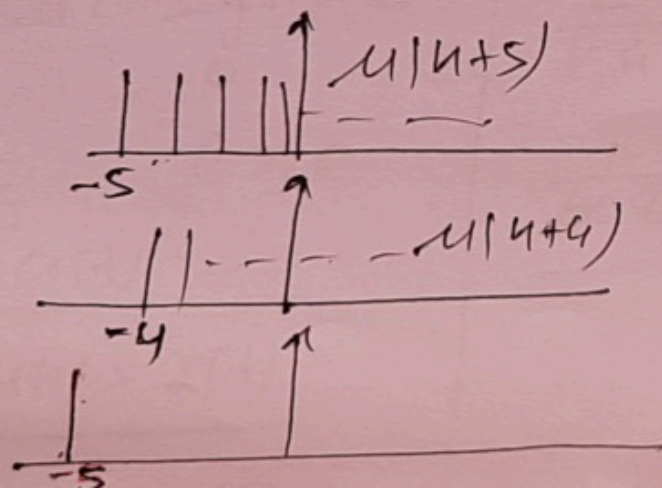


$$u(n+1) - u(n+6) = \underline{\delta(n+1)}$$



$$\underline{u(n-2) \otimes \delta(n+1) = u(n-2+1)} \\ = \underline{u(n+5)}$$

$$\underline{u(n+5) - u(n+4) = \delta(n+5)}$$



$$y(n) = \underline{f(n) \otimes \delta(n+5)}$$

$$y(n) = f(n+5)$$

$$f(n) = \{ \uparrow 2, 3 \}$$

$$f(n+5) = \{ 1, 2, 3, 0, 0, 0, \uparrow \} = y(n)$$



$$f(t) \otimes h(t)$$

# Problems on Convolution Method

①  $u(t) \otimes u(t)$

Time shifting Property

②  $\frac{e^{-at} u(t)}{f(t)} \otimes \frac{e^{-bt} u(t)}{h(t)}$

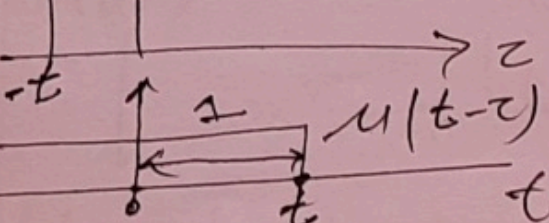
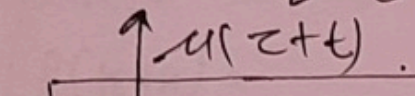
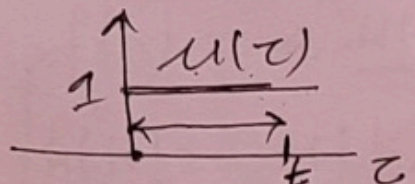
$$y(t) = \int_{-\infty}^{\infty} f(\tau) \cdot \underline{h(t-\tau)} \cdot d\tau$$

$$u(t-2) \otimes u(t-3) =$$

$$u(t-t_1) \otimes u(t-t_2) =$$

$$y(t) = \int_{-\infty}^{\infty} f(\tau) \cdot \underline{h(t-\tau)} \cdot d\tau$$

$u(\tau)$   ~~$h(\tau) \rightarrow h(t)$~~   
 $u(\tau) \rightarrow u(\tau+t)$



$$\int_0^t 1 \cdot 1 \cdot d\tau = \int_0^t 1 \cdot d\tau$$

$$x(t) = \frac{t}{2} u(t)$$

$$u(t) \otimes u(t) = x(t)$$

$$y(t) = \int_0^t e^{-a\tau} \cdot e^{-bt} \cdot e^{bz} \cdot d\tau$$

$$y(t) = e^{-bt} \int_0^t e^{(b-a)\tau} \cdot d\tau$$

$$= e^{-bt} \left[ \frac{e^{(b-a)\tau}}{(b-a)} \right]_0^t$$

$$\frac{e^{-bt}}{b-a} \left( e^{(b-a)t} - 1 \right)$$

$$\frac{1}{b-a} \left( e^{-at} - e^{-bt} \right) = y(t)$$

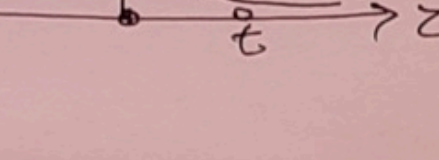
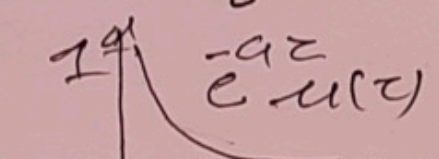
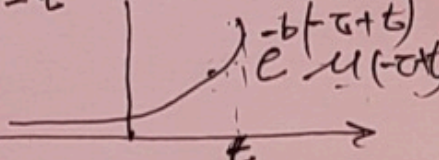
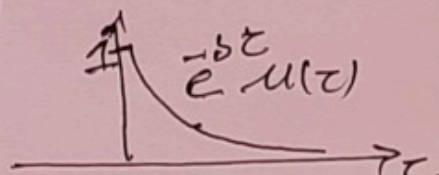
$$\frac{e^{-a\tau} u(\tau)}{f(\tau)}$$

~~$h(\tau) \rightarrow h(t)$~~   
 $h(\tau) \rightarrow h(\tau+t)$

$$h(-\tau+t)$$

$$h(\tau) = e^{-b\tau} u(\tau)$$

$$e^{-b(\tau+t)} u(\tau+t)$$





Q1

# Problems on Convolution Method

$$\textcircled{1} \frac{a^n u(n)}{f(n)} \otimes \frac{u(n)}{h(n)}$$

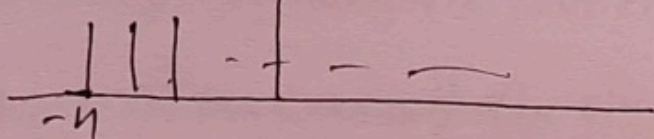
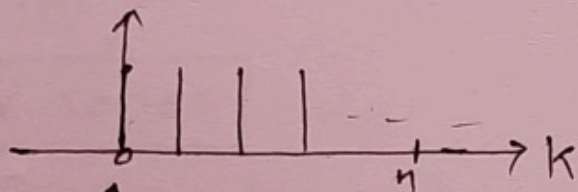
$$y(n) = \sum_{k=-\infty}^{\infty} \underline{f(k)} \cdot \underline{h(n-k)}$$

$$f(k) = a^k u(k) \quad k \geq 0$$

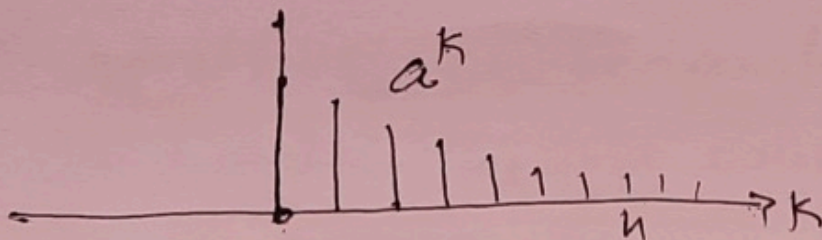
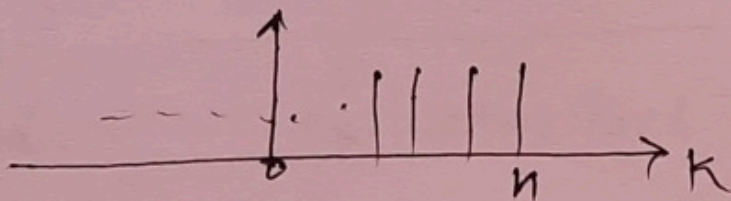
$$h(k) = u(k)$$

$$h(k+n) =$$

↓



$$h(-k+n) = \underline{h(n-k)}$$



$$y(n) = \sum_{k=0}^n a^k \cdot 1$$

$$= a^0 + a^1 + a^2 + a^3 + \dots + a^n$$

finite GP

$$y(n) = \frac{1-a^{n+1}}{1-a}$$